Optimisation of Food Grain Distribution in Gujarat:

Moving Wheat and Rice from FCI Depots to District Warehouses

CE 637 Infrastructure Systems: Planning and Management — Course Project

Abhitej Singh Bhullar (Roll: 23110009)
Radhika Purohit (Roll: 24110280)
Indian Institute of Technology Gandhinagar

November 2026

1. THE QUESTION AND THE MODEL

To ensure monthly NFSA food security across Gujarat, the **General Manager (Logistics & Movement), FCI Gujarat RO**, in coordination with the **Managing Director, GSCSCL**, must determine the optimal allocation of wheat and rice from 4 railhead depots to 6 district warehouses (132,000 MT total monthly demand) across 24 candidate highway corridors at minimum logistics and handling expenditure.

Linear Freight Optimization Model: For each route, freight cost scales linearly with road distance. Total system expenditure is minimized across all 48 shipment variables ($x_{ijk} \geq 0$):

$$\min Z = \sum_{i=1}^{4} \sum_{j=1}^{6} \sum_{k=1}^{2} c_{ij} x_{ijk}, \quad \text{where } c_{ij} = 100.00 + 2.50 \cdot d_{ij} \text{ [INR /MT]}$$

The allocation plan is constrained by 24 linear operational inequalities:

- **Meet district demand (12):** $\sum_{i=1}^{4} x_{ijk} \geq D_{jk} \quad \forall j,k$ (90,000 MT Wheat + 42,000 MT Rice = 132,000 MT)
- **Stay within depot stock (8):** $\sum_{j=1}^{6} x_{ijk} \leq S_{ik} \quad \forall i,k$ (Total Physical Stock: 112k MT Wheat + 55k MT Rice)
- **Depot handling capacity (4):** $\sum_{k=1}^{2} \sum_{j=1}^{6} x_{ijk} \leq C_i \quad \forall i$ (Shared truck bays & weighbridge limits)

Shared Handling Bottleneck. Depot dispatch capacity is tighter than physical stock across all depots ($C_i < S_{i1} + S_{i2}$). At FSD Chhani, 34,000 MT is stocked but only 28,000 MT can be loaded monthly, coupling wheat and rice into direct competition.

Algorithmic Solution & Telemetry. A custom Two-Phase Simplex solver was implemented with Bland's anti-cycling rule to clear initial demand infeasibility. The solver converged in exactly **56 pivots** (**Phase I: 28 pivots** to feasibility, **Phase II: 28 pivots** to optimality), matching SciPy HiGHS dual simplex (`highs-ds`) machine-precision cost (**Rs. 5.70 crore**).

2. THE LOWEST-COST DELIVERY PLAN

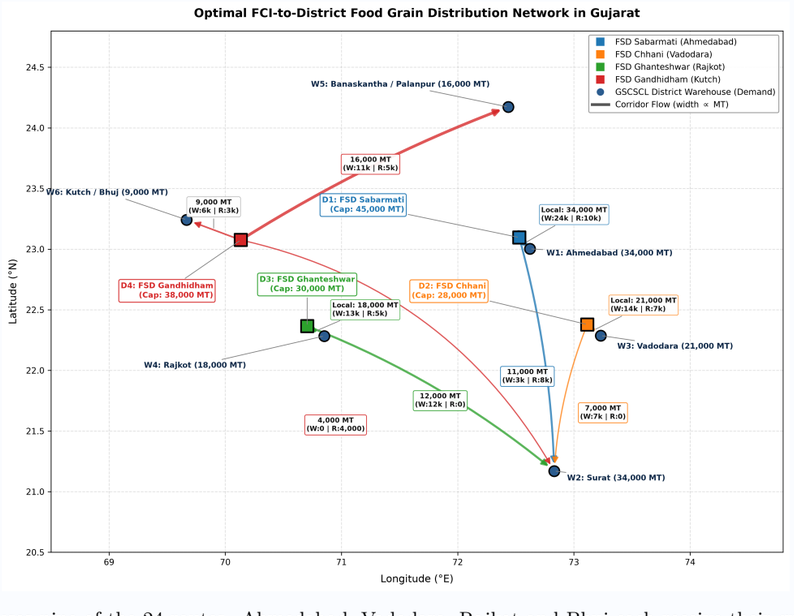
132,000 MT	Rs. 5.70 crore	INR 431.69	132.7 km
Grain delivered	Monthly cost	Cost per tonne	Average distance
90,000 wheat + 42,000 rice	INR 56,982,720.00	INR 43.17 per quintal	Weighted by tonnes sent

Monthly Shipments on the Nine Routes Used

Depot	District	Distance	Rs./MT	Wheat	Rice	Total MT
D1 Sabarmati	W1 Ahmedabad	6.7 km	116.85	24,000	10,000	**34,000**
D1 Sabarmati	W2 Surat	257.4 km	743.55	3,000	8,000	**11,000**
D2 Chhani	W2 Surat	154.9 km	487.20	7,000	0	**7,000**
D2 Chhani	W3 Vadodara	7.5 km	118.80	14,000	7,000	**21,000**
D3 Ghanteshwar	W2 Surat	429.8 km	1,174.48	12,000	0	**12,000**
D3 Ghanteshwar	W4 Rajkot	8.5 km	121.32	13,000	5,000	**18,000**
D4 Gandhidham	W2 Surat	564.0 km	1,510.10	0	4,000	**4,000**
D4 Gandhidham	W5 Banaskantha	320.3 km	900.70	11,000	5,000	**16,000**
D4 Gandhidham	W6 Kutch (Bhuj)	57.6 km	244.05	6,000	3,000	**9,000**
Monthly Total				**90,000**	**42,000**	**132,000**

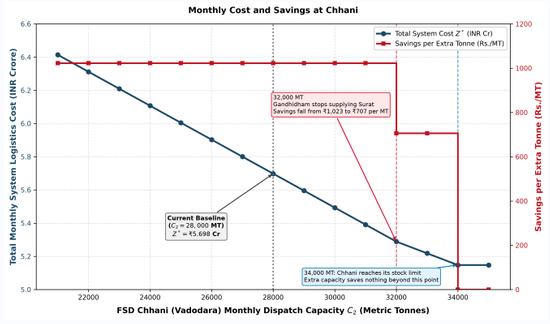
Why Surat needs grain from all four depots. Surat needs 34,000 MT and has no supply depot in this model. Chhani is about 155 km away. After sending 21,000 MT to Vadodara, it has room to load 7,000 MT for Surat. Sabarmati sends 11,000 MT, Ghanteshwar sends 12,000 MT and Gandhidham sends 4,000 MT from 564 km away.

3. WHERE THE GRAIN GOES



The lowest-cost plan uses nine of the 24 routes. Ahmedabad, Vadodara, Rajkot and Bhuj each receive their grain from a nearby depot. Surat receives grain from all four. The lines show which depots serve which districts. Thicker lines carry more grain.

4. WHAT IF CHHANI COULD LOAD MORE?



Monthly cost and savings from adding loading capacity at Chhani.

Two steps to lower monthly costs

- **Step 1 (28,000 to 32,000 MT).** Chhani replaces Gandhidham's shipments to Surat. Each tonne saves Rs. 1,510.10 minus Rs. 487.20, or **Rs. 1,022.90**. Over 4,000 MT, that saves **Rs. 40,91,600**.
- **Step 2 (32,000 to 34,000 MT).** Chhani takes over more of Surat's demand from Sabarmati, which can then replace Gandhidham in Banaskantha. The two changes save Rs. 256.35 and Rs. 450.20 per tonne, or **Rs. 706.55** together. Over 2,000 MT, that saves **Rs. 14,13,100**.
- Beyond 34,000 MT, more loading capacity saves nothing because Chhani has no more grain to send.

What FCI and GSCSCL Could Save

6,000 MT More Loading Capacity Saves Rs. 55.05 Lakh per Month

That is about Rs. 6.61 crore per year if demand and costs stay the same.